

Problem 1:

Model	Accuracy	Size
Int8 Quantization	0.43	5.8
Float16 Quantization	0.81	9.02
Dynamic Quantization	0.83	8.23
Pruned Model	0.54	14.12
Knowledge Distillation Student Model:	0.33	6.12

The model with the smallest size is the Int8 quantization. The Int8 had a size of 5.8KB, and Knowledge Distillation had the second smallest with 6.11KB. The model with the best performance is the Dynamic quantization, with an accuracy of 0.83.

A strategy to reduce the model size while maintaining strong classification performance was to combine two techniques of reducing the neural network architecture and applying TFLite quantization. The original model architecture was simplified by using smaller hidden layers with 32 and 16 neurons, which reduced the total number of trainable parameters. After training, Float16 quantization was applied to compress the model weights and lower the storage requirements.

The results were a TFLite size of 5.02KB, compared to 6.12KB for the original TFLite model. Despite the reduction in size, the model performed with 100% accuracy on the test dataset.

Quantization was the main factor in reducing model size, converting weights from 32-bit to 16-bit precision and significantly lowering memory usage with minimal impact on performance. Using smaller hidden layers also helped by reducing the number of parameters.

Further reductions (such as INT8 quantization or pruning) are possible, but the model already achieves 100% accuracy, so additional compression may risk degrading performance. Overall, the current model provides a strong balance between compact size and classification accuracy.

Problem 2: Exploring Edge Impulse

1. Classification, Regression, Anomaly Detection (GMM), Anomaly Detection (K-means), Anomaly Detection. Use classification for categorizing movement or recognizing audio, regression for predicting numeric continuous values, GMM for finding unlikely events

based on comparison to the normal distribution, K-means for finding unlikely events based on distance to clusters, and Anomaly Detection for finding unlikely events based on deviation from other inputs.

2. Dense, which is a fully connected layer, and the simplest form of a neural network layer. 1D Convolution / pooling, where the learn features take spatial information into account along a single dimension, and 2D convolution / pooling, which does the same as 1D but along two dimensions. Reshape, which turns one-dimensional data from a DSP block into multi-dimensional data. Flatten, which flattens multi-dimensional data into a single dimension. Lastly is Dropout, which reduces the risk of a model overfitting the dataset by randomly cutting a fraction of network connections during training.
3. Under the deployment section there is quantized, which is int8, and unoptimized, which is float32.
4. What input modalities are supported for synthetic data generation in the free-tier version. For each modality, list the synthetic data generation methods/algorithms provided. Explain why synthetic data is important in machine learning applications. The two options are image and speech generation. There are two options for images: Black Forest Labs Flux-Pro Synthetic Image Generator, and DALL-E 3 Synthetic Image Generator. For the speech generator it is Whisper Synthetic Voice Generator. Using synthetic data helps balance datasets which improves model robustness by making more diverse data. Machine Learning needs a lot of data, which sometimes there is not enough data available. Synthetic data helps make more data available to train the model.